EXP 6

Customer Segmentation using K-Means and Hierarchical Clustering

1. Dataset Source

The dataset used for this experiment is obtained from **Kaggle**.

Dataset Link:
https://www.kaggle.com/datasets/vjchoudhary7/customer-segmentation-tutorial-in-python

Dataset Name: **Customer Segmentation Dataset**

2. Dataset Description

The **Customer Segmentation Dataset** contains information about customers of a shopping mall. The dataset is commonly used to demonstrate clustering algorithms for market segmentation.

Dataset Size

- Total Records: **200 customers**
- Total Features: **5 columns**

Features Description

Feature	Description
CustomerID	Unique identifier assigned to each customer
Gender	Gender of the customer (Male/Female)
Age	Age of the customer
Annual Income (k$)	Annual income of the customer in thousand dollars
Spending Score (1–100)	Score assigned by the mall based on customer spending behavior

Target Variable

This dataset **does not contain a predefined target variable** because clustering is an **unsupervised learning technique**. The goal is to group customers based on similar characteristics.

Dataset Characteristics

- Contains both **categorical and numerical features**
- Requires **data preprocessing**
- Useful for **customer segmentation**
- Small dataset suitable for demonstrating clustering techniques

3. Mathematical Formulation of the Algorithms

3.1 K-Means Clustering

K-Means is an unsupervised learning algorithm used to partition a dataset into **K clusters**.

Objective Function

The algorithm minimizes the **within-cluster sum of squares (WCSS)**:

$$J = \sum_{i=1}^{k} \sum_{x \in C_i} ||x - \mu_i||^2$$

Where:

- kk = number of clusters
- C_iCi = set of data points in cluster ii
- μ_iμi = centroid of cluster ii
- $||x-\mu_i||^2$||x−μi||2 = Euclidean distance between point and centroid

Algorithm Steps

1. Choose number of clusters KK
2. Randomly initialize cluster centroids
3. Assign each data point to the nearest centroid
4. Recalculate centroids
5. Repeat steps 3–4 until convergence

3.2 Hierarchical Clustering

Hierarchical clustering builds a **tree-like structure (dendrogram)** representing nested clusters.

Two types:

- **Agglomerative (bottom-up)**
- **Divisive (top-down)**

Distance Calculation

Distance between clusters is calculated using linkage methods.

Example: **Single Linkage**

$$D(A, B) = \min_{x \in A, y \in B} d(x, y)$$

Where:

- A, BA,B = clusters
- $d(x, y)$d(x,y) = distance between points

Other linkage methods include:

- Complete linkage
- Average linkage
- Ward linkage

Agglomerative Process

1. Treat each point as a cluster
2. Compute pairwise distances
3. Merge closest clusters
4. Update distance matrix
5. Repeat until one cluster remains

4. Algorithm Limitations

K-Means Limitations

1. **Requires predefined K value**
2. Sensitive to **initial centroid selection**
3. Works best with **spherical clusters**
4. Sensitive to **outliers**
5. Not suitable for **non-linear cluster shapes**

Hierarchical Clustering Limitations

1. **High computational cost** for large datasets
2. Once clusters are merged **cannot be undone**
3. Memory intensive
4. Performance depends heavily on **distance metric and linkage method**

5. Methodology / Workflow

The following steps were performed in the experiment.

Step 1: Data Collection

Download dataset from Kaggle.

Step 2: Data Preprocessing

- Import dataset using **Pandas**
- Check for missing values
- Encode categorical variables (Gender)
- Select relevant features:
    - Annual Income
    - Spending Score

Step 3: Exploratory Data Analysis

- Visualize distribution of features
- Plot scatter plots to observe patterns

Step 4: Feature Scaling

Apply **StandardScaler** or **Normalization** to standardize values.

Step 5: K-Means Clustering

- Use **Elbow Method** to determine optimal KK
- Train K-Means model
- Assign cluster labels

Step 6: Hierarchical Clustering

- Compute distance matrix
- Generate **dendrogram**
- Choose cluster cutoff
- Assign clusters

Step 7: Visualization

Plot clusters using scatter plots for interpretation.

Workflow Diagram

Dataset Collection
↓
Data Preprocessing
↓
Feature Selection
↓
Feature Scaling
↓
K-Means Clustering
↓
Hierarchical Clustering
↓
Visualization & Interpretation
↓
Performance Analysis

6. Performance Analysis

Since clustering is **unsupervised**, evaluation is done using internal metrics.

1. Silhouette Score

Measures how similar a data point is to its cluster compared to other clusters.

$$S = \frac{b - a}{max(a, b)}$$

Where:

- aa = average intra-cluster distance
- bb = average nearest-cluster distance

Silhouette score ranges from **-1 to 1**

- **1 → well clustered**
- **0 → overlapping clusters**
- **-1 → incorrect clustering**

2. Elbow Method

Used to determine optimal number of clusters by plotting:

Number of Clusters vs WCSS

The point where WCSS starts decreasing slowly is the **optimal K**.

Results Interpretation

- Customers were segmented into **distinct groups based on spending behavior and income**
- K-Means produced clear clusters
- Hierarchical clustering provided a **visual dendrogram structure**

Typical clusters found:

- High income – high spending
- High income – low spending
- Low income – high spending
- Low income – low spending

7. Hyperparameter Tuning

Hyperparameter tuning improves clustering performance.

K-Means Hyperparameters

Parameter	Description
n_clusters	Number of clusters
init	Initialization method (k-means++)
max_iter	Maximum number of iterations
n_init	Number of centroid initializations

Tuning Process

1. Use **Elbow Method** to select optimal clusters.
2. Experiment with different **n_clusters values (2–10)**.
3. Choose cluster number with **best silhouette score**.

Example:

Clusters	Silhouette Score
3	0.41
4	0.48
5	0.55
6	0.50

Best value: **K = 5**

Hierarchical Clustering Hyperparameters

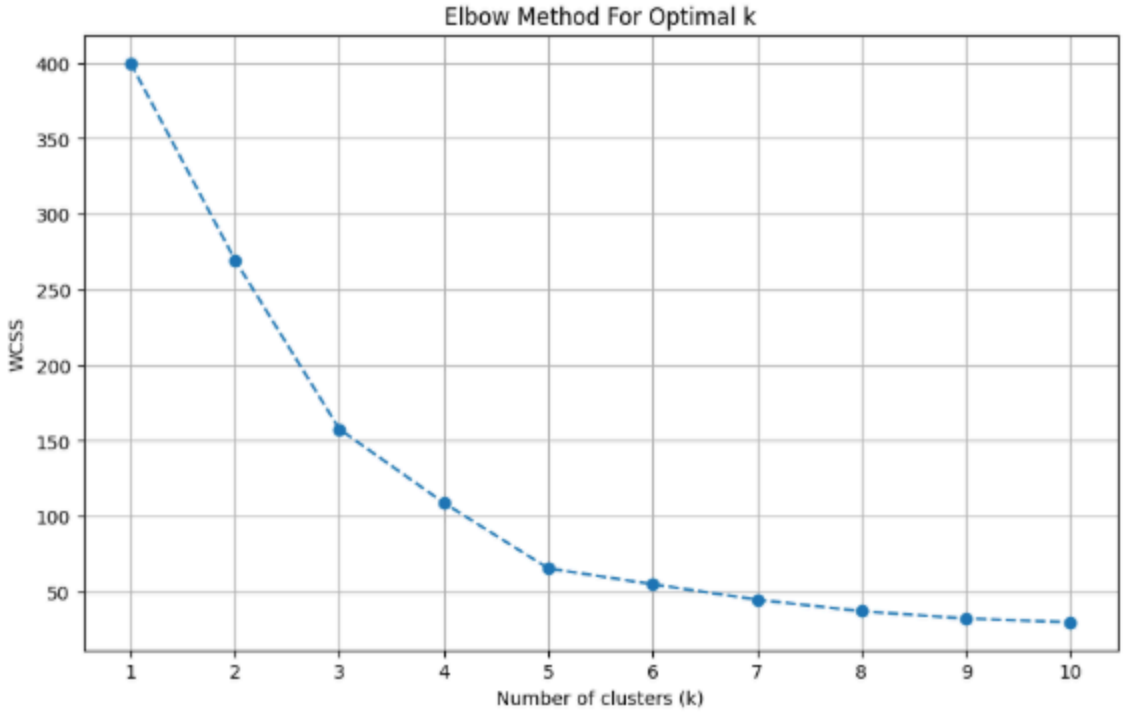
Parameter	Description
linkage	Distance strategy (ward, single, complete)
affinity	Distance metric (euclidean, manhattan)

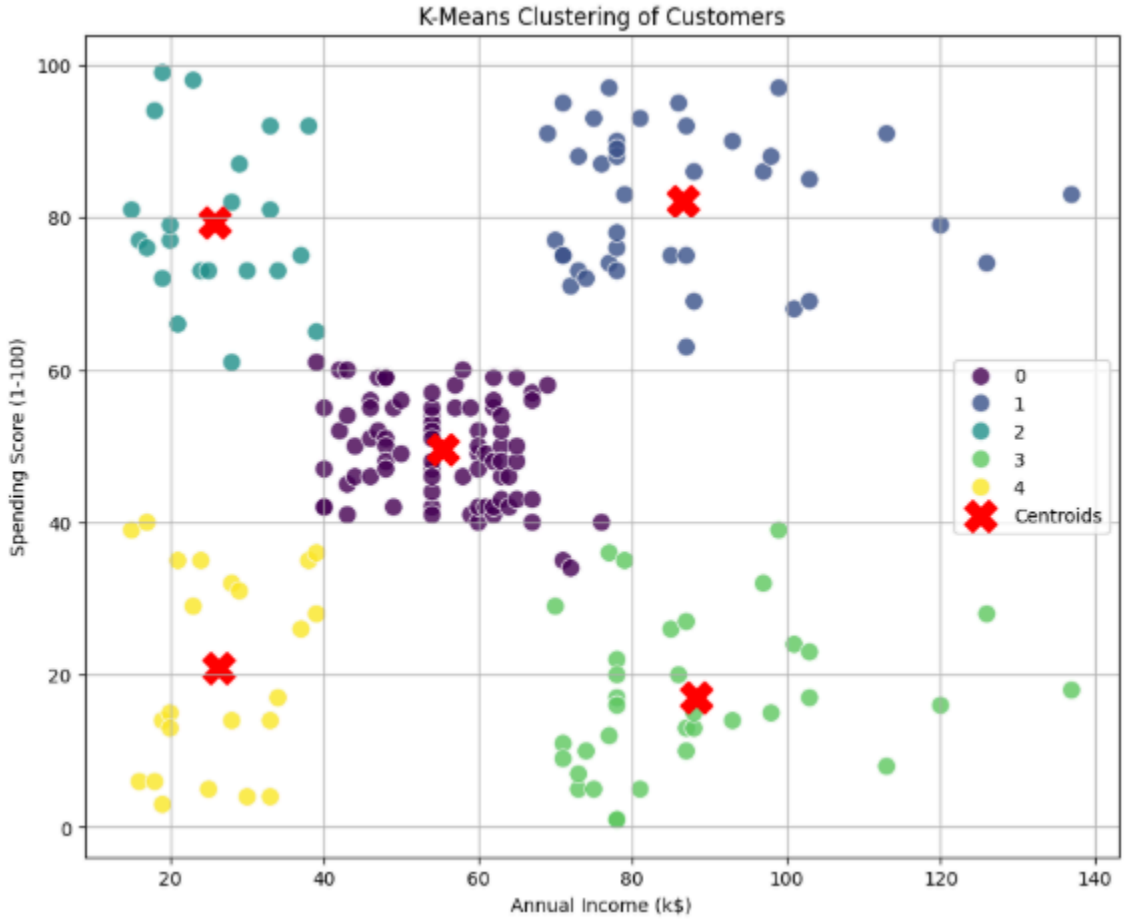
Tuning Strategy

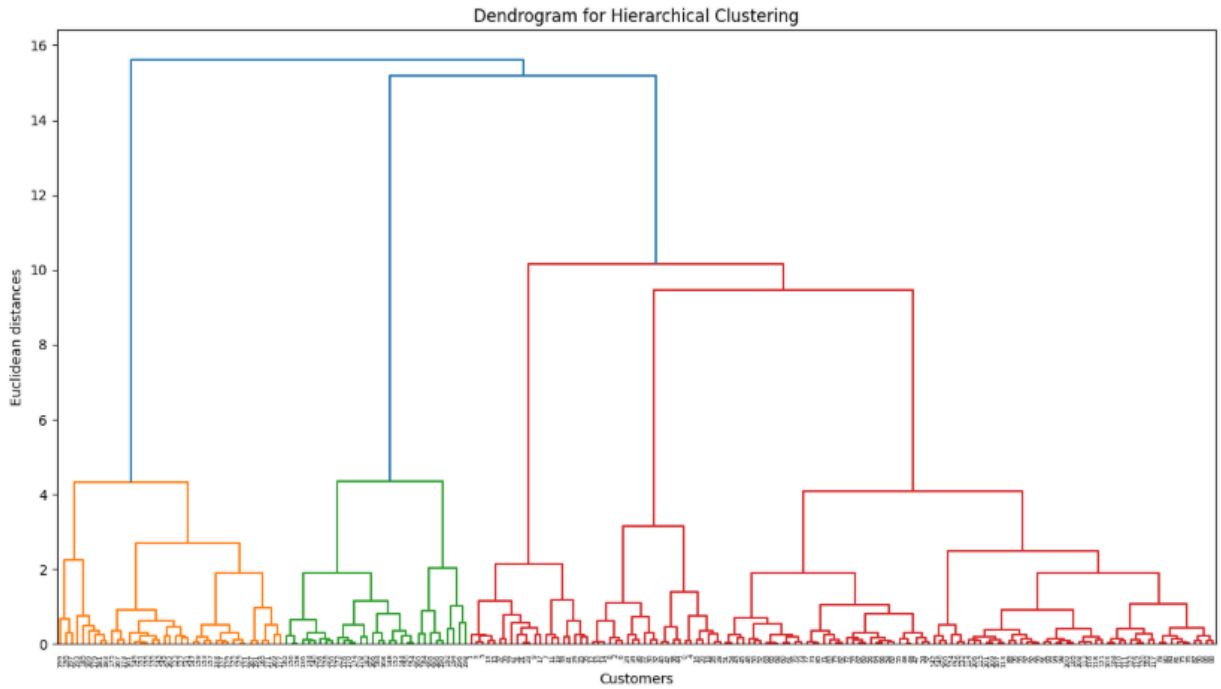
- Test different linkage methods
- Evaluate cluster separation
- Compare with silhouette score

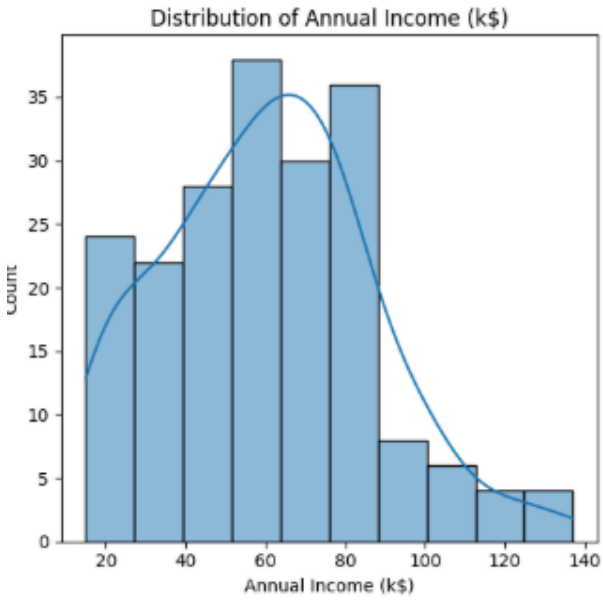
Best results are typically achieved using **Ward linkage with Euclidean distance**.

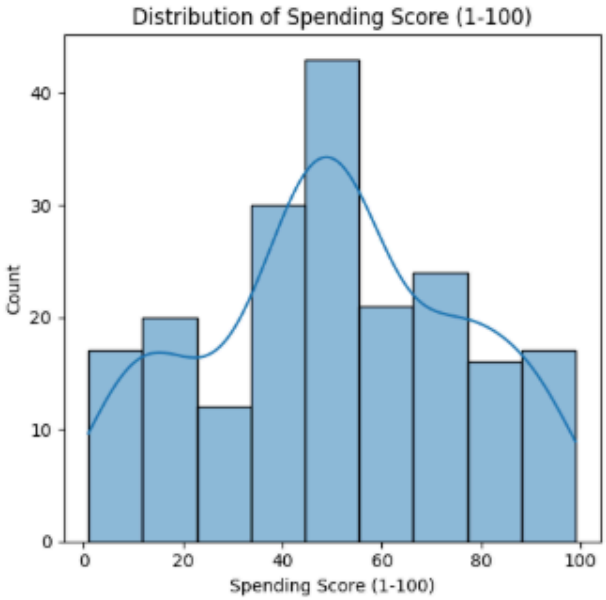
	CustomerID	Gender	Age	Annual Income (k$)	Spending Score (1-100)	KMeans_Cluster	HC_Cluster
0	1	Male	19	15	39	4	4
1	2	Male	21	15	81	2	3
2	3	Female	20	16	6	4	4
3	4	Female	23	16	77	2	3
4	5	Female	31	17	40	4	4


Elbow Method For Optimal k
WCSS
Number of clusters (k)
400
350
300
250
200
150
100
50
1
2
3
4
5
6
7
8
9
10


K-Means Clustering of Customers
Spending Score (1-100)
Annual Income (k$)
0
1
2
3
4
Centroids
100
80
60
40
20
0
20
40
60
80
100
120
140







Conclusion

In this experiment, **K-Means and Hierarchical Clustering** were applied to a customer segmentation dataset. The algorithms successfully grouped customers based on income and spending patterns. The experiment demonstrated how unsupervised learning can be used for **market segmentation and business insights**.